Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

Cancer. Author manuscript; available in PMC 2020 May 21.

Published in final edited form as:

Cancer. 2013 May 01; 119(9): 1736–1743. doi:10.1002/cncr.27915.

Phase II Study of Pre-Excision Single Dose Intraoperative Radiation Therapy (IORT) for Early Stage Breast Cancers: Six year update with application of the ASTRO Accelerated Partial Breast Irradiation Consensus Statement Criteria

Noam A. VanderWalde, M.D.[a,b], **Ellen L. Jones, M.D. PhD**[a,b], **Randall J. Kimple, M.D. PhD**[c], **Dominic T. Moore, M.P.H**[b], **Nancy Klauber-DeMore, M.D**[b,d], **Carolyn I. Sartor, M.D.**[a,b], **David W. Ollila, M.D.**[b,d,*]

[a]Department of Radiation Oncology, University of North Carolina at Chapel Hill, Chapel Hill, NC

[b]Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, NC

[c]Department of Radiation Oncology, University of Wisconsin School of Medicine, Madison, WI

[d]Department of Surgery, University of North Carolina at Chapel Hill, Chapel Hill, NC

Abstract

Background: Intraoperative Radiation Therapy (IORT) allows delivery of high dose radiation at the time of lumpectomy, potentially sparing adjuvant daily radiation. We performed a Phase II study of pre-excision IORT for early-stage breast cancer.

Methods and Materials: Patients ≥ 48 with invasive ductal carcinoma, ≤ 3cm, and clinically node negative were eligible on this IRB-approved study. Ultrasound was used to select electron energy and cone size to cover the tumor plus 1.5 - 2.0 cm lateral margins and 1 cm deep margins (90% isodose). Fifteen Gy was delivered with a Mobetron irradiator. Immediate needle localized partial mastectomy followed. Local event results were updated using Kaplan-Meier method.

Results: 53 patients received IORT alone. Median age was 63. Median tumor size was 1.2 cm. 81% were ER or PR+, 11% were Her2+ and 15% were triple negative. 42%, 49%, and 9% would have fallen into the Suitable, Cautionary and Unsuitable Groups of the ASTRO consensus statement for APBI respectively. Median follow up was 69 months. Ipsilateral events occurred in 8 of 53 patients. The 6-year actuarial rate of ipsilateral events was 15% (95% CI 7-29%). The crude event rate for Suitable and Cautionary Groups was 1/22 (5%) and 7/26 (27%) respectively. Overall survival was 94.4%, and breast cancer specific survival was 100%.

*Corresponding Author: David W. Ollila M.D.,Department of Surgery, Division of Surgical Oncology and Endocrine Surgery, University of North Carolina at Chapel Hill, 101 Manning Drive, Chapel Hill, NC 27599, Phone: (919) 966-5221, Fax: (919) 966-8806, david_ollila@med.unc.edu.

Financial Disclosures: There are no financial disclosures from any of the authors.

Informed Consent: All patients who participated in this IRB approved study signed consent prior to accrual on this study.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Conclusions: Our rate of local events is concerning, especially in the Cautionary group. Based on these findings, pre-excision IORT, as delivered in this study, may not provide adequate local control for less favorable early-stage breast cancers.

Condensed Abstract:

Between 2003-2007, we performed a Phase II study to test the efficacy and toxicity of pre-excision intraoperative radiation therapy for early stage breast cancer patients. Of the patients who received this therapy without adjuvant local therapy, 15% had ipsilateral breast recurrences.

Keywords

Breast Neoplasm; Radiation Therapy; Intraoperative Procedure; Radiotherapy Dosage; Surgery

Introduction:

There is increasing interest among physicians and patients in shortening the course and amount of adjuvant radiation in early stage breast cancer. Current standard of care treatment options include mastectomy or breast conserving surgery followed by daily whole breast radiation over a period of 3.5[1] to 7 weeks[2,3] sometimes coordinated with systemic cytotoxic and/or hormonal therapy. The daily fractions of radiation often lead to inconvenience for patients and may lead women to avoid their adjuvant radiation.[4,5] In addition, some estimate that 75% of local in-breast recurrences occur in the quadrant of the previous tumor.[6]

There have been multiple studies over the last decade[7,8] evaluating the use of Accelerated Partial Breast Irradiation (APBI) as an alternative to standard fractionated whole breast radiotherapy. There is an ongoing large multi-institutional study (NSABPB-39/RTOG 0413) evaluating multi-catheter brachytherapy, MammoSite, or external beam APBI versus whole breast radiation. However, a modality left out of that important study is Intraoperative Radiation Therapy (IORT). Intraoperative Radiation Therapy consists of a large single dose of radiation therapy, most often delivered to the tumor bed immediately following surgical excision while the patient is still under anesthesia in the operating room. IORT has the potential of making APBI even more convenient for patients by dramatically reducing the treatment time and discomfort. IORT was originally reported as a substitute for the delivery of a lumpectomy cavity boost dose following whole breast irradiation.[9]

IORT as a boost to the tumor bed prior to adjuvant whole breast radiation was first piloted by Vaidya et al. in 1998, using their TARGIT kilovoltage photon delivery system.[10] Around the same time, a group of researchers in Milan began using a mobile linear accelerator to deliver 21 Gy of electrons intraoperatively to the tumor cavity as a substitute for whole breast irradiation.[11] Since that time there have been at least three other groups studying the use of different methods of delivering post-lumpectomy IORT as an alternative to whole breast irradiation.[12–14] In 2003, our institution initiated a prospective phase II trial of pre-excisional IORT as a novel way to deliver APBI. It was felt that delivery of IORT to the intact tumor would help overcome problems with target coverage/localization and improve the radiation dose distribution. The technical details, cosmesis/toxicity results, and 3 year local control, from this study have previously been reported.[15–17] We update the local event

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

data, analyzing both patient and treatment characteristics of those who received IORT alone. Further, we retrospectively considered the differences in outcomes based on subgroups defined by the 2009 American Society of Therapeutic Radiation Oncology (ASTRO) consensus statement for the appropriateness of APBI.[18]

Methods and Materials:

Patient Selection:

The technical details of patient selection, IORT and surgical technique were described previously.[15–17] Briefly, patients age ≥ 48 with invasive ductal carcinomas ≤ 3cm by mammogram and ultrasound, with sonographically visible tumors, and clinically negative lymph nodes were eligible for treatment on this IRB-approved Phase II trial. All patients were required to sign informed consent for participation. Patients with ductal carcinoma in situ, invasive lobular carcinoma, multicentric disease, skin or chest wall involvement, bilateral breast cancer, or who received neoadjuvant chemotherapy or irradiation to the involved breast were excluded. The primary endpoints of the study were rates of good or excellent cosmesis and toxicity. These were previously reported by Kimple et al.[15] The secondary clinical endpoint was ipsilateral breast recurrence.

Treatment Technique:

Prior to entering the operating room, an ultrasound simulation for radiation planning was performed in diagnostic radiology during the needle localization wire placement. The ultrasound was used to determine the best angle of approach in order to minimize the skin-to-tumor distance, and maximize the tumor-to-lung distance. Then, an electron cone was chosen to encompass the tumor with a 1.5–2.0 cm radial margin. These margins were chosen based on pathological data suggesting that only 15% of patients have tumor greater than 2 cm from the reference mass.[19] The distance from the skin to the anterior tumor edge, the posterior tumor edge, and the pleural surface were measured. Then, electron energy sufficient to deliver 15 Gy to the intact tumor plus a 1.0 cm deep margin to the 90% isodose line was chosen (Figure 1a).

The patients were taken to the operating room, where sentinel lymphadenectomy was performed. Following confirmation of negative nodes on frozen section, the skin over the needle localization wire was incised and the radiation target was identified. The radiation cone was positioned at the angle determined by the ultrasound simulation just abutting the breast tissue so as not to compress the tissue. The radiation field was positioned over the region of the tumor, as determined by visualization and palpation (Figure 1b). A mobile electron accelerator (Mobetron, Intraop Medical, Norcross, GA) was used to deliver the electron radiation to the intact tumor (Figure 1c). Following the radiation, a standard needle localized partial mastectomy was performed with a gross clear margin of removed normal tissue. The excised tissue was then sent for pathology.

Postoperative Local/Systemic Therapy:

Recommendations for additional local therapy were made, per protocol, based on pathological review. Patients with positive margins or close margins (<1 mm) were

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

recommended re-excision to negative margins or mastectomy. Patients with extensive intraductal component, microscopic tumor >3cm, or invasive lobular carcinoma were recommended for either whole breast radiotherapy (without a boost) or mastectomy. Patients with positive sentinel lymph nodes on final pathology were recommended for axillary lymph node dissection and either mastectomy or whole breast radiation and therefore were not considered in this cohort. Systemic hormonal therapy and chemotherapy were recommended at the discretion of medical oncology, and were not specified in the protocol.

Follow up:

Patients were evaluated for complications and recurrence with history and physical exams at 1 week, 3, 6, and 12 months post-RT, every 6 months years 2-5 post-RT, and annually thereafter. Mammograms with ultrasounds were performed every 6 months for the first 5 years, then annually thereafter.

Statistical Methods:

The Kaplan-Meier method was used to calculate local control. Fisher's exact test was used to test for differences in general association between recurrence groups for categorical variables. The nonparametric Jonckheere-Terpstra method was used to test for ordered differences over categories of interest (described below). The Wilcoxon rank-sum test (using Van der Waerden normal scores) was used for two-group comparisons for continuous variables. While statistical comparisons were made through statistical testing, not unexpectedly, no statistically significant differences were demonstrated between the recurrence and non-recurrence groups. The lack of statistical significance could be due in large part to the lack of statistical power (due to the small number of recurrences, n=8). Therefore, appropriate descriptive statistics have been reported, and p-values for the not significant tests have been omitted. Statistical analyses were performed using SAS, Version 9.2 (SAS Institute, Inc., Cary, NC).

Results:

There were 71 patients who received IORT on our phase II study from March 2003 to July 2007. Of those 71 patients, 18 received further local therapy due to 'high risk' pathology. Eleven patients received whole breast RT and 7 received mastectomy. In this current report, we only analyzed the 53 patients who received IORT without further local therapy. The baseline characteristics of the 53 patients who had IORT alone and their median treatment characteristics were indicative of a low risk population (Table 1). Based on the 2009 ASTRO Consensus Statement on Accelerated Partial Breast Irradiation (APBI), 22 patients were Suitable, 26 Cautionary, and 5 Unsuitable.[18] As of July, 2012 median follow up was 69 months (range 3 – 98 months).

There have been a total of 8 ipsilateral breast events for a crude rate of 15.1% (5 patients had true recurrences within the prior lumpectomy cavity, 1 patient had a DCIS within the lumpectomy site, 1 patient had a multicentric recurrence with masses within the lumpectomy site and elsewhere in the ipsilateral breast, and 1 patient had local recurrence within the lumpectomy site, regional axillary recurrence, and distant recurrence with metastatic disease

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

in the lung and bones). The 5 and 6 year actuarial rates of local events are 13% (95% CI 6–26%) and 15% (95% CI 7–29%), respectively (Figure 2). Six of the patients were Estrogen receptor positive (ER+), Progesterone receptor positive (PR+) and Human Epidermal Growth Factor Receptor 2 negative (Her2 –), and 2 were "triple negative". All six hormone positive patients were recommended hormonal blockade. The triple negative patients both received adjuvant chemotherapy. When we retrospectively applied the 2009 ASTRO APBI Consensus statement to our entire study cohort and then specifically to patients with ipsilateral breast events we found; one patient was within the 'Suitable' group, 7 were in the 'Cautionary' group, and none were in the Unsuitable group. Thus, the local event rates were 1/22 (5%) in the Suitable, 7/26 (27%) in the Cautionary, and 0/5 (0%) in the Unsuitable. Although the percentage difference in recurrence rates appeared to suggest a clinical trend towards higher recurrence rates among the 'Cautionary' group, this was not statistically significant when compared to those who recurred in the 'Suitable' group (p= 0.11) (Figure 3).

Five of the local events have been treated with repeat partial mastectomy followed by whole breast radiation. One patient underwent a mastectomy and a negative sentinel lymph node biopsy. The patient with multicentric disease was treated with mastectomy and post-mastectomy radiation therapy to the chest wall. The above 7 patients are currently without further evidence of disease. However, the patient with local, regional, and distant failure underwent a modified radical mastectomy and axillary lymph node dissection and is now undergoing chemotherapy. When we analyzed patient, tumor (i.e. size, grade, hormone status, etc.), and treatment characteristics (cone size, electron energy, dose, ultrasound characteristics, etc.), we found no major differences between the 8 patients who developed local events and the 45 who did not (Table 2). One patient in the IORT alone group developed lung cancer 3.5 years after treatment and died of her lung cancer. Two additional patients died of non-cancer related conditions. The group's crude overall survival was 94.3%, and breast cancer specific survival was 100%.

Discussion:

Our 6 year actuarial local event rate of 15% was unexpected. The Early Breast Cancer Trialists' Collaborative Group Meta-analysis[20] estimated a 5 year local recurrence rate of 7.2% for patients receiving excision plus whole breast RT. However, in our population of patients with small primary tumors, node negative disease, and mostly hormone positive patients, we would expect a 5 year risk of recurrence following excision plus whole breast RT closer to 3.5%[1].

To date, there have been five other studies of IORT alone for low risk breast cancer patients (Table 3). In a review of prospectively accrued patients performed by the Milan group,[21] 1822 patients with low risk cancers received excision plus intraoperative radiotherapy (ELIOT) alone. The patients in this study were low risk, however roughly 30% of their patients would have either been ineligible for our study or would have required adjuvant local therapy based on node positive disease or positive margins. An additional 20% would not have been eligible due to histology (i.e. lobular carcinoma/DCIS). They delivered the intraoperative radiation *following* quandractectomy using one of two mobile linear

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

accelerators (Novac7 and Liac) to deliver 21 Gy with electrons to the 90% isodose line. In their recent updated manuscript they had a total of 74 events in 1797 evaluable patients (4.1%).[22] Importantly, the purpose of that article was to evaluate the appropriateness of the ASTRO APBI consensus statement to single dose IORT. They reported 5 year in-breast recurrence rates (events/person years x 5 years, assuming equal events per year) of 1.5%, 4.4%, and 8.8% for the 'Suitable', 'Cautionary', and 'Unsuitable' groups respectively (p= .003). The Milan group has also initiated a prospective randomized trial using the same (ELIOT) technique, but to date no data has been presented from this study.

The second and only published randomized trial of IORT alone versus standard fractionated whole breast radiotherapy is the TARGIT A trial.[23] This phase III international trial enrolled 2232 patients; 1113 patients on the IORT arm. Of those 1113 patients, 854 patients received IORT without further local therapy with outcomes performed as an 'intention-to-treat' analysis. They had strict post-treatment histopathologic criteria for recommending further local therapy. However, 19% of the patients in their IORT arm had positive lymph nodes. They delivered their intraoperative radiation following lumpectomy using the Intrabeam device to deliver 20 Gy with 50 kV x-rays to the surface of the lumpectomy cavity with attenuation to 5–7 Gy at 1 cm depth. With a median follow up of roughly 25 months, their ipsilateral breast recurrence rate among their IORT cohort was 0.5% (6/1113).

The study from Memorial Sloan Kettering[24] enrolled 52 women with early stage breast cancer to receive post-excision intraoperative radiation with their institutionally-designed H.A.M. (Harrison-Anderson-Mick) applicator. The system used a remotely after-loaded Ir^{192} source to deliver 18–20 Gy to 1.0 cm margins in all directions except deep, which was prescribed to 0.5 cm. Of their 52 patients, only 4 would not have qualified for our study due to unfavorable histopathologic findings at lumpectomy. With a median follow up of 68 months, their local failure rate was 7.7% (4/52).

The Montpellier group enrolled 96 women above the age of 65 for their phase II study.[14] They used a dedicated linear accelerator located in the operating room to deliver 21 Gy electrons to the 90% isodose line to the tumor cavity. Applicator tubes were placed within the tumor cavity prior to delivery of radiation. Similar to our study they had strict pathological criteria and 29 patients did not receive IORT secondary to upstaging of tumor size or node positive disease found on frozen section. Of the 42 patients who received IORT, 2 had positive margins and received mastectomies, and 9 had either positive nodes (6), lobular carcinoma (2), or multifocal disease (1). None of those 9 patients received adjuvant local therapy. With 30 months of median follow up, they reported 2 breast ipsilateral breast recurrences among their 40 patients who received IORT alone (5%).

The last study, enrolled 226 patients in Verona, Italy.[13] They used a Mobetron accelerator to deliver 21 Gy electrons to Dmax (or 16.8 Gy to 80% isodose line) to a re-approximated tumor cavity. Approximately 22 % of their patients had positive nodes and 15% had close or positive margins. All patients with positive margins had re-excision, but neither the close margin or positive node patients underwent further whole breast radiotherapy. With a mean follow up of 46 months, they only observed one ipsilateral breast event, which occurred outside of the treated quadrant. Their ipsilateral recurrence rate was 0.4%.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

The most notable difference between the above studies and our trial is the IORT technique. We are the only group that delivered the intraoperative RT *prior* to excision. The theoretical benefit to delivering radiation to in-situ tumor is that the radiation may result in a more uniform dose distribution of the tumor and surrounding tissues. When IORT is delivered post-excision, there are potential challenges to assuring a relative uniform dose distribution to the targeted structures, related in-part to heterogeneities in the cavity walls and tissue folds. Three of the above studies added material to the excision cavity to address this issue. In the two Italian studies, the excision cavity walls were re-approximated during radiation. When the trial was designed it was felt that delivering IORT pre-excision would allow better tumor localization and better dose distribution.

A major disadvantage of delivering IORT prior to excision is that the margins of tissue irradiation were determined by a pre-operative ultrasound, and we did not mandate intraoperative ultrasound assessment. Therefore, the intraoperative tumor margins may have been underestimated or the tumor may not have been completely centered within the electron field leading to suboptimal coverage. Further, the targeted tissues were only 1.5 – 2 cm around the radiographic abnormality. This may also have provided suboptimal coverage of microscopic spread. These issues may have contributed to our higher-than-expected local event rate. With the increasing interest in the use of preoperative partial breast irradiation[25] we recommend caution be used in designing pre-operative treatment volumes.

The second major difference in our study is dose. We delivered a median dose of 15 Gy with electrons to the 90% isodose line. With the exception of the TARGIT A trial, the other studies delivered a dose of 20-21 Gy to either 90% isodose line, Dmax or at a depth of 1 cm. The TARGIT A trial delivered 20 Gy to the surface of the cavity, but due to the use of low energy (50kV) x-rays, the dose at 1 cm was closer to 5-7 Gy. The lower dose in our trial was originally chosen in order to minimize toxicity especially at the deep pleuritic margin, thus allowing us to avoid shielding devices, such as those used in the Milan, and Verona studies. If one were to perform a biologically effective dose (BED) calculation using the equation $BED = D [1+ d (\alpha/\beta)]$ using an α/β of 4 for breast cancer one would find a BED of 71.3 Gy_4 for a dose of 15Gy x 1 which is similar to the 75 Gy_4 BED of 2 Gy x 25 fraction course. However, the linear-quadratic model was not designed to calculate BED for single large doses, and therefore may overestimate the true BED to the tumor.[26] Thus it is possible that we under-dosed these tumors leading to our higher-than-expected local event rate.

Our trial and the MSK trial have the highest recurrence rates (15% and 7.7% respectively) and also appear to have the longest median follow up (69 months and 68 months) . However, our two studies also had far fewer patients compared to the Milan, Verona, and TARGIT A studies, and thus, the less favorable outcomes could be due to statistical variation or a manifestation of the ‘learning curve when new techniques are introduced clinically.

Patient selection for APBI may be critical as well. Our study was closed before the publication of the ASTRO APBI consensus statement in 2009. Therefore, our inclusion/exclusion criteria did not match the ASTRO groups, with the most notable differences of age ≥ 48 and lack of consideration for lymphovascular space invasion. However, just like the Milan study, we had a higher rate of recurrence in our patients who were retrospectively

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

considered in the 'Cautionary' group.[18] However this was not statistically significant. Our lack of statistical significance was likely due to the small number of patients in the Unsuitable category (n=5) and their lack of recurrences. Given our small number of events and patients, we cannot determine the true cause of the increased rate of recurrence for the 'Cautionary' group. However, we feel our rates of recurrence in the 'Cautionary' group (27%) support using the ASTRO APBI criteria in the selection of patients for IORT.

In conclusion, due to our high ipsilateral breast event rate, we do not recommend the use of IORT delivered to the pre-excision tumor at a dose of 15 Gy alone using the margins and localization technique described above, especially in patients in the ASTRO consensus statement Cautionary group. The use of IORT, as delivered in other trials remains a potential option, but is best performed in the setting of a prospective clinical trial. Long term follow up of the larger clinical series will yield important information regarding efficacy and toxicity of these IORT techniques in comparison to standard fractionated whole breast radiotherapy.

Acknowledgments

Funding: This study was supported by UNC/Lineberger Comprehensive Cancer Center, Sisko, and the Rodney and Ruth James Foundation.

References

1. Whelan TJ, Pignol JP, Levine MN, et al.: Long-term results of hypofractionated radiation therapy for breast cancer. N Engl J Med 362:513–20, 2010 [PubMed: 20147717]
2. Fisher B, Anderson S, Bryant J, et al.: Twenty-year follow-up of a randomized trial comparing total mastectomy, lumpectomy, and lumpectomy plus irradiation for the treatment of invasive breast cancer. N Engl J Med 347:1233–41, 2002 [PubMed: 12393820]
3. Bartelink H, Horiot JC, Poortmans PM, et al.: Impact of a higher radiation dose on local control and survival in breast-conserving therapy of early breast cancer: 10-year results of the randomized boost versus no boost EORTC 22881–10882 trial. J Clin Oncol 25:3259–65, 2007 [PubMed: 17577015]
4. Lazovich DA, White E, Thomas DB, et al.: Underutilization of breast-conserving surgery and radiation therapy among women with stage I or II breast cancer. JAMA 266:3433–8, 1991 [PubMed: 1688350]
5. Athas WF, Adams-Cameron M, Hunt WC, et al.: Travel distance to radiation therapy and receipt of radiotherapy following breast-conserving surgery. J Natl Cancer Inst 92:269–71, 2000 [PubMed: 10655446]
6. Gujral DM, Sumo G, Owen JR, et al.: Ipsilateral breast tumor relapse: local recurrence versus new primary tumor and the effect of whole-breast radiotherapy on the rate of new primaries. Int J Radiat Oncol Biol Phys 79:19–25, 2011 [PubMed: 20471183]
7. Shah C, Antonucci JV, Wilkinson JB, et al.: Twelve-year clinical outcomes and patterns of failure with accelerated partial breast irradiation versus whole-breast irradiation: results of a matched-pair analysis. Radiother Oncol 100:210–4, 2011 [PubMed: 21497927]
8. Polgar C, Fodor J, Major T, et al.: Breast-conserving treatment with partial or whole breast irradiation for low-risk invasive breast carcinoma−-5-year results of a randomized trial. Int J Radiat Oncol Biol Phys 69:694–702, 2007 [PubMed: 17531400]
9. Dobelbower RR, Merrick HW, Eltaki A, et al.: Intraoperative electron beam therapy and external photon beam therapy with lumpectomy as primary treatment for early breast cancer. Ann Radiol (Paris) 32:497–501, 1989 [PubMed: 2517384]
10. Vaidya JS, Baum M, Tobias JS, et al.: Targeted intra-operative radiotherapy (Targit): an innovative method of treatment for early breast cancer. Ann Oncol 12:1075–80, 2001 [PubMed: 11583188]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

11. Veronesi U, Orecchia R, Luini A, et al.: Full-dose intraoperative radiotherapy with electrons during breast-conserving surgery: experience with 590 cases. Ann Surg 242:101–6, 2005 [PubMed: 15973107]
12. Sacchini V, Beal K, Goldberg J, et al.: Study of quadrant high-dose intraoperative radiation therapy for early-stage breast cancer. Br J Surg 95:1105–10, 2008 [PubMed: 18690634]
13. Maluta S, Dall'oglio S, Marciai N, et al.: Accelerated Partial Breast Irradiation Using Only Intraoperative Electron Radiation Therapy in Early Stage Breast Cancer. Int J Radiat Oncol Biol Phys, 2012
14. Lemanski C, Azria D, Gourgon-Bourgade S, et al.: Intraoperative radiotherapy in early-stage breast cancer: results of the montpellier phase II trial. Int J Radiat Oncol Biol Phys 76:698–703, 2010 [PubMed: 19467579]
15. Kimple RJ, Klauber-DeMore N, Kuzmiak CM, et al.: Cosmetic outcomes for accelerated partial breast irradiation before surgical excision of early-stage breast cancer using single-dose intraoperative radiotherapy. Int J Radiat Oncol Biol Phys 79:400–7, 2011 [PubMed: 20395062]
16. Kimple RJ, Klauber-Demore N, Kuzmiak CM, et al.: Local Control Following Single-Dose Intraoperative Radiotherapy Prior to Surgical Excision of Early-Stage Breast Cancer. Ann Surg Oncol, 2010
17. Ollila DW, Klauber-DeMore N, Tesche LJ, et al.: Feasibility of breast preserving therapy with single fraction in situ radiotherapy delivered intraoperatively. Ann Surg Oncol 14:660–9, 2007 [PubMed: 17091330]
18. Smith BD, Arthur DW, Buchholz TA, et al.: Accelerated partial breast irradiation consensus statement from the American Society for Radiation Oncology (ASTRO). Int J Radiat Oncol Biol Phys 74:987–1001, 2009 [PubMed: 19545784]
19. Holland R, Veling SH, Mravunac M, et al.: Histologic multifocality of Tis, T1–2 breast carcinomas. Implications for clinical trials of breast-conserving surgery. Cancer 56:979–90, 1985 [PubMed: 2990668]
20. Clarke M, Collins R, Darby S, et al.: Effects of radiotherapy and of differences in the extent of surgery for early breast cancer on local recurrence and 15-year survival: an overview of the randomised trials. Lancet 366:2087–106, 2005 [PubMed: 16360786]
21. Veronesi U, Orecchia R, Luini A, et al.: Intraoperative radiotherapy during breast conserving surgery: a study on 1,822 cases treated with electrons. Breast Cancer Res Treat 124:141–51, 2010 [PubMed: 20711810]
22. Leonardi MC, Maisonneuve P, Mastropasqua MG, et al.: How Do the ASTRO Consensus Statement Guidelines for the Application of Accelerated Partial Breast Irradiation Fit Intraoperative Radiotherapy? A Retrospective Analysis of Patients Treated at the European Institute of Oncology. Int J Radiat Oncol Biol Phys 83:806–13, 2012 [PubMed: 22245196]
23. Vaidya JS, Joseph DJ, Tobias JS, et al.: Targeted intraoperative radiotherapy versus whole breast radiotherapy for breast cancer (TARGIT-A trial): an international, prospective, randomised, non-inferiority phase 3 trial. Lancet 376:91–102, 2010 [PubMed: 20570343]
24. Beal K, Sacchini V, Zelefsky M, et al.: Five Year Update on Intraoperative Radiation Therapy for Breast Cancer, ASTRO 2011. Miami, FL, Int J Radiat Oncol Biol Phys, 2011, pp S241–S242
25. Bartelink H, Bourgier C, Elkhuizen P: Has partial breast irradiation by IORT or brachytherapy been prematurely introduced into the clinic? Radiother Oncol, 2012
26. Kirkpatrick JP, Meyer JJ, Marks LB: The linear-quadratic model is inappropriate to model high dose per fraction effects in radiosurgery. Semin Radiat Oncol 18:240–3, 2008 [PubMed: 18725110]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

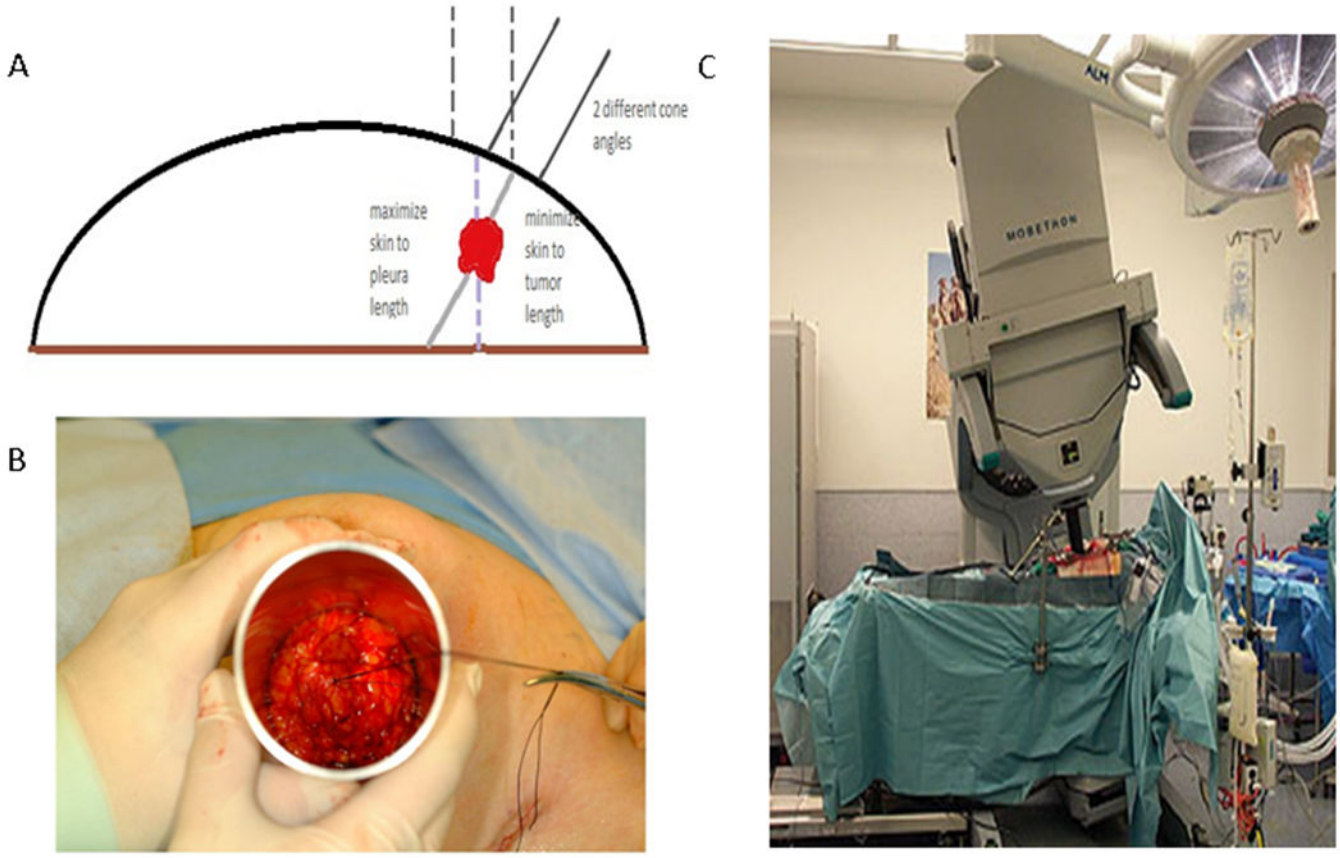

Figure 1:

Figure 1a: Ultrasound simulation was performed to aid in tumor localization and for optimal angle of approach to minimize the skin-to-tumor distance and maximize the tumor- to-lung distance.

Figure 1b: The skin was retracted and the cone was placed overlying the tumor at the angle of approach per Ultrasound simulation. The tumor was centered inside the cone and visualized by both the treating radiation oncologist and surgeon.

Figure 1c: The patients were positioned underneath the mobile linear accelerator which was positioned at the correct angle for delivery of 15 Gy to the 90% isodose line.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

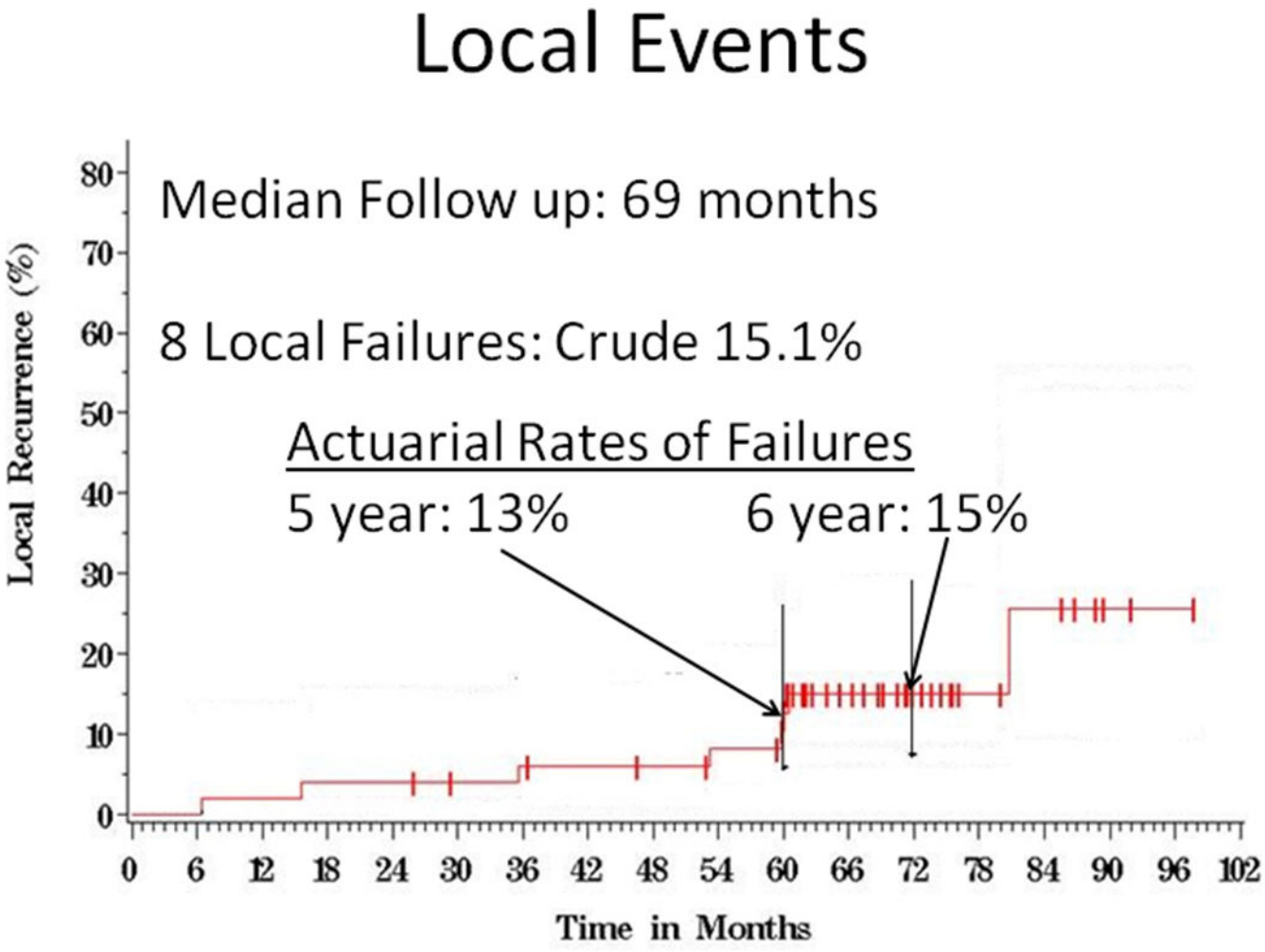

Figure 2:

Time to Local Event Graph with local recurrence % on y axis and time in months on the x-axis. The red tick marks are indicators of patient lost to follow up. The black lines at 5 years and 6 years represent the 95% confidence intervals at those times.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

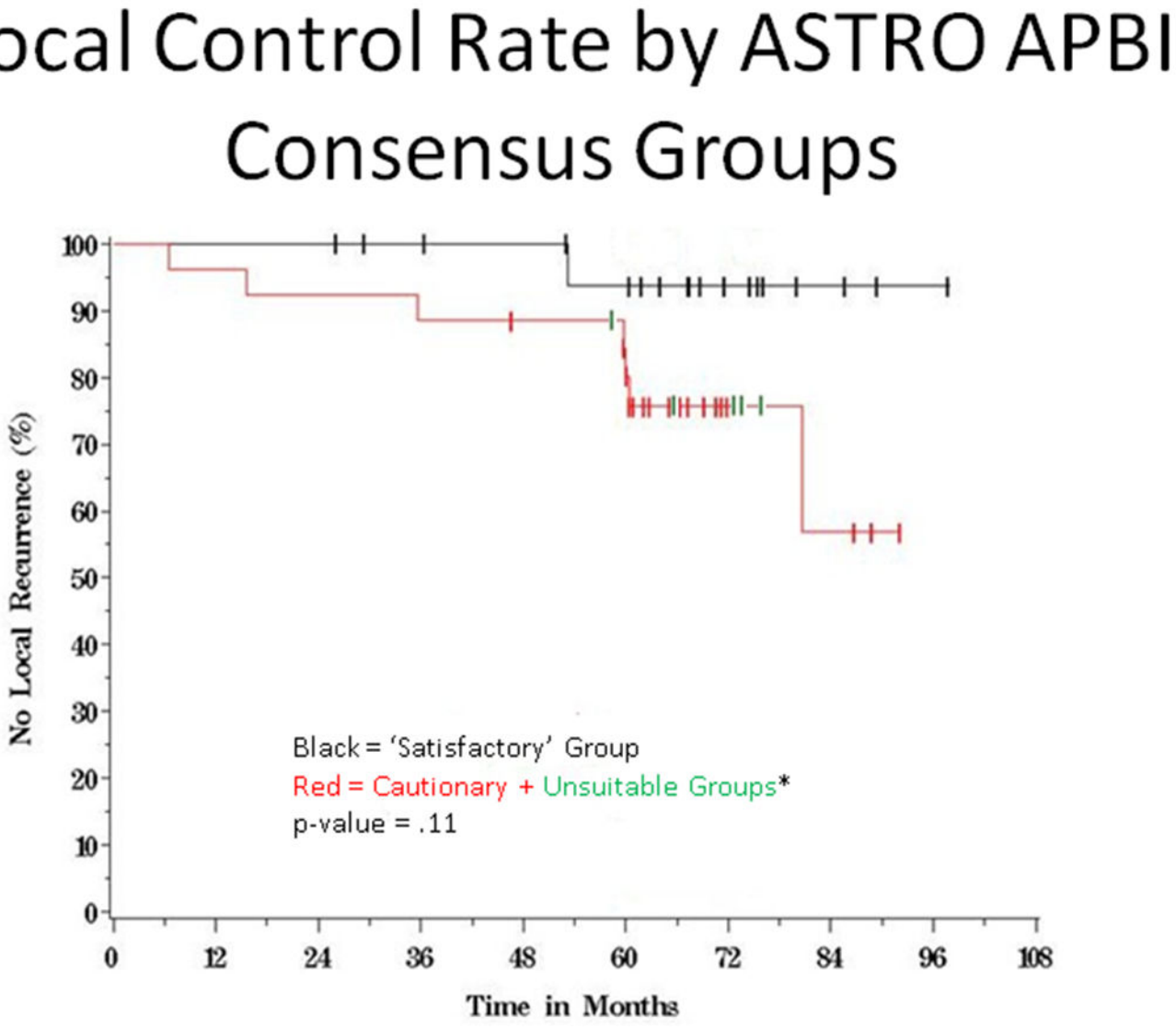

Figure 3:

Kaplan-Meier Local Control Rate by ASTRO APBI Consensus Groups. The Black line indicates the local control rate for 'Suitable' patients and tick marks indicate individual patients lost to follow up. The red line indicates the local control rate for 'Cautionary' patients with red tick marks indicating individual patients lost to follow up. Because there were 0 of 5 'Unsuitable patients' who recurred there is no Green line, however the Green Tick marks indicate individual 'Unsuitable' patients lost to follow up.

*Note there were no recurrences in the Unsuitable group and 7 recurrences in the Cautionary group, however these two groups were combined for clinical and statistical purposes. Green tick marks indicate last follow up for each Unsuitable patient

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 1:

Patient and Treatment Characteristics

Treatment Characteristics n=53	**Median**	**Range**
Tumor size on US	1.1	0.5-2.38
Skin to anterior tumor depth	1	0-2.6
Skin to posterior tumor depth	2.14	0-5.1
skin to pleura depth	3.9	1.57-6.7
Dose dmax	15	14.3-23.7
Path tumor size	1.2	0.3-2.7
Difference between US size and path size	0.195	range −0.7-2.0
Cone size	5.5	4.0 - 7.0
electron energy	12	6.0-12.0
Age	63	48-84
Patient Characteristics	**n = 53**	**%**
Pathological stage		
T1a	2	4
T1b	12	23
T1c	30	56
T2 ≤ 3 cm	9	17
G1	20	38
G2	20	38
G3	13	24
Hormone Status		
ER or PR+	43	81
Her2 +	6	11
Triple Negative	8	15
ASTRO APBI groups		
Suitable	22	42
Cautionary	26	49
Unsuitable	5	9
Systemic Treatment		
Endocrine Alone	24	49
Chemo Alone	4	8
Both	4	8

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 2:

Comparison of Characteristics between Patients with Local Events and those Without

Treatment Characteristics	**Local Events n=8 Median (Range)**	**No Local Events n=45 Median (Range)**
Tumor size on US (cm)	0.97 (0.6-2.3)	1.15 (0.5-2.38)
Skin to anterior tumor depth (cm)	0.7 (0-2.4)	1.15 (0-2.6)
Skin to posterior tumor depth (cm)	2.27 (0.9-4.4)	2.1 (0-5.1)
skin to pleura depth (cm)	4.0 (3.4-6.7)	3.85 (1.57-5.0)
Dose (Gy)	15.13 (15-19)	15 (14.34-23.73)
Path tumor size (cm)	1.6 (0.7-2.6)	1.2 (0.3-2.7)
Difference between US size and path size (cm)	0.14 (−0.5-2.0)	0.2 (−0.7-1.56)
Cone size (cm)	5 (4.5 - 6.0)	5.5 (4.0-7.0)
electron energy (MeV)	12 (9.0-12.0)	12 (6.0-12.0)
Age (years)	62.5 (55-82)	63.0 (48-84)
Patient Characteristics	**Recurrence n=8 (%)**	**No Recurrence n=45 (%)**
Pathological stage		
T1a	0	4
T1b	25	27
T1c	63	56
T2 ≤ 3 cm	13	18
G1	38	38
G2	38	38
G3	25	27
Hormone Status		
ER or PR+	75	82
Her2 +	0	13
Triple Negative	25	13
ASTRO APBI groups		
Suitable	12	47
Cautionary	88	42
Unsuitable	0	11
Systemic Treatment		
Endocrine Alone	25	49
Chemo Alone	13	7
Both	0	9

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 3:

Comparison of 6 IORT studies

IORT studies	Type of Study	# of IORT patients	Median F/U (months)	Type/Dose of Radiation	Ipsilateral Breast Event Rates of IORT Alone
LCCC 0218	Phase II	53	69	**Pre-Lumpectomy**: Electrons-**15** Gy	15.1%
TARGIT A Trial[23]	Phase III	1113	25-26 *	Post Lumpectomy: kV x-rays-20 Gy	0.5%
Milan Study[22]	Retrospective	1822	60 ǂ	Post-Lumpectomy: Electrons-21 Gy	4.1% ǂ
MSKCC Study[24]	Pilot	52	68	Post-Lumpectomy: Ir192-20 Gy	7.7%
Montpellier Study[14]	Phase II	42	30	Post-Lumpectomy: Electrons-21 Gy	5.0% ¥
Verona Study[13]	Phase II	226	46 (mean)	Post-Lumpectomy: Electrons-21 Gy	0.4%

* Median follow up not clearly stated in publication but derived from numbers in Kaplan Meier curve

ǂ Failure rate not explicitly stated in publication, derived by total number of failures (74) divided by patients evaluated in publication (1797)

ǂ Median follow up not stated in publication, however 5 year rates of recurrence given, therefore assume at least 5 years of median follow up.

¥ Derived by total number of failures (2) by number of patients who received IORT alone (40).